

Lab 04 - Modbus/TCP Emulation and Register Access



Prof. Dr. Luiz F. Freitas-Gutierrez

luiz.gutierrez@ufsm.br

linkedin.com/in/lffreitas-gutierrez



Lattes

ORCID

SciProfiles

Scopus

ResearcherID

substationworm

LFreitasGutierrez

Tasks

- **1** After starting OTLab 04, access the `otlab-student` workstation via VNC at `10.1.0.40:5901` using the password `123456`. Hint: if you are running OTLab 04 on a Linux system, it is recommended to use Remmina to access the graphical interface of the `otlab-student` workstation. If you are using Windows with WSL, the graphical interface may be accessed through `localhost:5901`, provided that WSL is running in WSL2 mode and no firewall restrictions are present.
- **2** Within the graphical interface of the `otlab-student` workstation accessed through VNC, add a Modbus slave in ModbusPal, a Java-based Modbus simulator: slave ID: 1, slave name: Slave.
- **3** Edit the newly created Modbus slave by adding five holding registers with addresses 1 to 5 and assigning the corresponding values 10, 20, 30, 40, and 50.
- **4** After completing these configurations, click **Run** in ModbusPal and open a terminal on the `otlab-student` workstation (`./OTLab04.sh -run`).
- **5** Verify the IP address of the `otlab-student` workstation.
- **6** Determine the subnet range of the network where the `otlab-student` workstation is deployed.
- **7** Discover the IP address, MAC address, and vendor information of the `otlab-student` workstation.
- **8** Use `favalex/modbus-cli` to read the holding registers of the slave configured in ModbusPal. Hint: `modbus -s <Slave_ID> <IPAddress> 0 1 2 3 4` (where addresses 0 to 4 correspond to the first five holding registers).

Tools

The following tools are available on the `otlab-student` workstation to complete OTLab 04: `ifconfig`, `masscan`, `netdiscover`, `nmap`, and `modbus`.

Nomenclature

- ID: identifier.
- IP: Internet Protocol.
- MAC: Media Access Control.
- TCP: Transmission Control Protocol.
- UDP: User Datagram Protocol.
- VNC: Virtual Network Computing.
- WSL: Windows Subsystem for Linux.